The q-Borel plus q-Laplace image of the q-Hermite-Weber fundamental solution equals the generalized mu-function

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned journal source file sources/sigma-2023-014-a-generalization-of-zwegers-mu-function-acco/pinned-original-source.tex (A Generalization of Zwegers' mu-Function According to the q-Hermite-Weber Difference Equation); no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: Seed convention: counters stepped from the source itself (tmp-step-2023 walk, cross-validated against the host aux ground truth of the pinned source compiled whole): section 1 of the shared [thm] counter holds one environment (def:mua 1.1) before this theorem, and section 1 holds numbered equations (1.1)-(1.21) before the span, so the card prints Theorem 1.2 with in-span equation (1.22) eq:mu and q-Hermite–Weber (label line 379). The proof is the author's own 'Proof of Theorem' environment (source lines 746-780) delivered verbatim including the optional-argument back-reference to the theorem and the tag-qed plus renewcommand-qed device; all its displays are unnumbered (align* and displayed equations), so the card carries no further numbered equations after (1.22). The statement sits in Section 1 and the proof is delivered in Section 2 of the article; since the proof carries no numbered equations, the single-seed card numbering reproduces the article's printed numbers exactly. Out-of-span navigation pointers: none (no \ref outside the counted spans). Citations: the proof carries cite GR once (line 747, q-binomial theorem p. 8 equation (1.3.2)); resolved by the retained-bibliography recipe (duty ruling 2026-10-14, per Chief instruction to lane-x2, 022 precedent L48545): the entry is copied verbatim from the author's own thebibliography (source lines 1531-1533) into the card-local fragment retained-bibliography.tex with bibliography backend none.

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Editorial scope. the counted statement and its own proof environment only

Allowed standard prerequisites (editorial).

- The series definition of the generalized mu-function def:mua (1.1, source lines 343-362), the value the proof identifies at the end.
- The fundamental solution of the q-Hermite-Weber equation as the 2phi0 series t-tilde-f_0 (source lines 365-369) and the q-Laplace transformation eq:q-Laplace (1.21, source line 331) composing it into f_0.
- The q-binomial theorem, cited at line 747 from entry GR (p. 8, equation (1.3.2)), a standard published input used to sum the q-Borel image.

Theorem 1.2. *Let $f_0(x,\lambda)$ be the image of the q-Borel and q-Laplace transformations of the fundamental solution $\widetilde{f}_0$ of the q-Hermite–Weber equation at $x=0$:*

$$f_0(x) := x^{\frac{\alpha}{2}}\mathcal{L}^+ \circ \mathcal{B}^+\big(\widetilde{f}_0\big)(x,\lambda).$$

Then we have

$$f_0\big(\mathrm{e}^{2\pi\mathrm{i}(u-v)}, -\mathrm{e}^{2\pi\mathrm{i}u}\big) = \mu(u,v;\alpha). \tag{1.22}$$

Proof of Theorem 1.2. From the definition of the q-Borel transformation and the q-binomial theorem (see [1, p. 8, equation (1.3.2)]),

$$\mathcal{B}^+\left(\widetilde{f}_0\right)(\xi) = \sum_{n\geq 0}\frac{(a)_n}{(q)_n}(-1)^n\left(\frac{\xi}{a}\right)^n = \frac{(-\xi)_\infty}{(-\xi/a)_\infty}.$$

By a simple calculation, we have

$$\begin{aligned}\mathcal{L}^+ \circ \mathcal{B}^+\big(\widetilde{f}_0\big)(x,-\lambda) &= \sum_{n\in\mathbb{Z}}\frac{1}{\theta_q(-\lambda q^n/x)}\frac{(\lambda q^n)_\infty}{(\lambda q^n/a)_\infty}\\ &= \frac{1}{\theta_q(-\lambda/x)}\sum_{n\in\mathbb{Z}}\left(-\frac{\lambda}{x}\right)^{n+1} q^{\frac{n(n+1)}{2}}\frac{\big(\lambda q^{n+1}\big)_\infty}{\big(\lambda q^{n-\alpha+1}\big)_\infty}.\end{aligned}$$

The second equality follows from the well-known q-difference relation of the theta function:

$$\theta_q(q^n x) = q^{-\frac{n(n-1)}{2}} x^{-n} \theta_q(x), \qquad n \in \mathbb{Z}.$$

By substituting $x = \mathrm{e}^{2\pi \mathrm{i}(u-v)}$ and $\lambda = \mathrm{e}^{2\pi \mathrm{i} u}$, we have

$$\begin{aligned} f_0\big(\mathrm{e}^{2\pi \mathrm{i}(u-v)}, -\mathrm{e}^{2\pi \mathrm{i} u}\big) &= \mathrm{e}^{\pi \mathrm{i}\alpha(u-v)} \mathcal{L}^+ \circ \mathcal{B}^+\big(\widetilde{f_0}\big)\big(\mathrm{e}^{2\pi \mathrm{i}(u-v)}, -\mathrm{e}^{2\pi \mathrm{i} v}\big) \\ &= -\frac{\mathrm{e}^{\pi \mathrm{i}\alpha(u-v)}}{\theta_q\big(-\mathrm{e}^{2\pi \mathrm{i} v}\big)} \sum_{n\in\mathbb{Z}} (-1)^n \mathrm{e}^{2\pi \mathrm{i}(n+1)v} q^{\frac{n(n+1)}{2}} \frac{\big(\mathrm{e}^{2\pi \mathrm{i} u} q^{n+1}\big)_\infty}{\big(\mathrm{e}^{2\pi \mathrm{i} u} q^{n-\alpha+1}\big)_\infty} \\ &= \frac{\mathrm{i}\mathrm{e}^{\pi \mathrm{i}\alpha(u-v)}}{\vartheta_{11}(v)} \sum_{n\in\mathbb{Z}} (-1)^n \mathrm{e}^{2\pi \mathrm{i}\left(n+\frac{1}{2}\right)v} q^{\frac{\left(n+\frac{1}{2}\right)^2}{2}} \frac{\big(\mathrm{e}^{2\pi \mathrm{i} u} q^{n+1}\big)_\infty}{\big(\mathrm{e}^{2\pi \mathrm{i} u} q^{n-\alpha+1}\big)_\infty} \\ &= \mathrm{i} q^{\frac{1}{8}} \mu(u, v; \alpha). \end{aligned}$$

□

References

[1] Gasper G., Rahman M., Basic hypergeometric series, 2nd ed., *Encyclopedia Math. Appl.*, Vol. 96, Cambridge University Press, Cambridge, 2004.